

Qatar-1b

R-0626

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_230921 · created 2026-07-25T13:53:09+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460208.7469 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.168 +/- 0.021
Transit depth (Rp/Rs)^2	2.81 +/- 0.72 [%]
Orbital Inclination (inc)	82.43 +/- 1.01
Airmass coefficient 1 (a1)	1.175 +/- 0.016
Airmass coefficient 2 (a2)	-0.117 +/- 0.017
Scatter in the residuals of the lightcurve fit is	1.38 %
Best Comparison Star	None
Optimal Aperture	2.74
Optimal Annulus	9.12
Transit Duration (day)	0.0595 +/- 0.0092

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.168 $\pm$ 0.021 vs 0.1463 (deviation 0.87 σ)
Duration fit vs published	0.0595 d vs 0.0692 d (deviation 0.88 σ)
Mid-time O-C	-6.7 min
Depth SNR	6.7
Residual scatter	1.38 %

Light curve

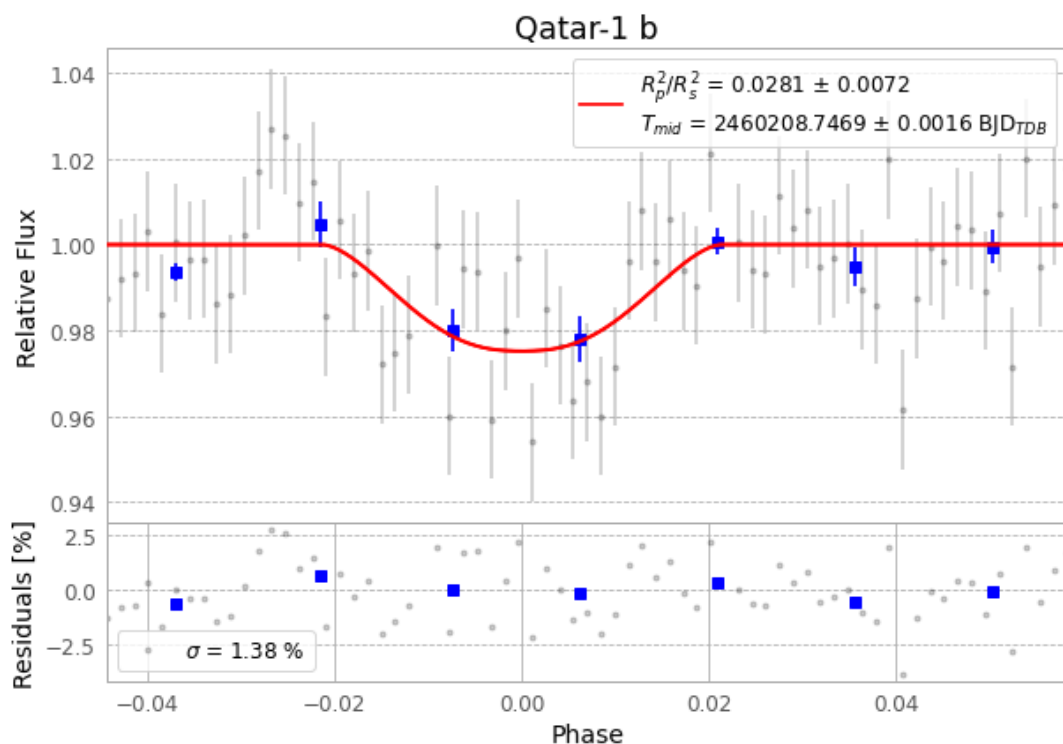


Figure 1 — FinalLightCurve_Qatar-1 b_2023-09-20.png (SHA-256 98a4721c687540c8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [273, 191], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2e4e8ffd680baccf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

71 FITS frames (47 MB), Qatar-1230921042115.FITS → Qatar-1230921075115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-230921.log (e3a91e5278d0...), FinalLightCurve_Qatar-1 b_2023-09-20.png (98a4721c6875...), FinalLightCurve_Qatar-1 b_2023-09-20.csv (afad58f4ef69...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 13:53Z.